

Biodegradable polymer as controlled release system of organic antifoulant to prevent marine biofouling

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ARTICLE INFO

Article history:

Received 28 September 2016

Received in revised form 7 November 2016

Accepted 12 December 2016

Available online 22 December 2016

Keywords:

Degradation

Controlled release

Biofouling

Polyurethane

ABSTRACT

Controlled release of antifoulant is critical for marine antibiofouling. We have prepared polyurethane with degradable polyester segments consisting of poly(ethylene adipate) (PEA), poly(1,4-butylen adipate) (PBA) or poly(1,6-hexamethylene adipate) (PHA) and used it as the release system for organic antifoulants. The degradation rate of the polyurethane increases with the content of the degradable segments. On the other hand, as the segments change from PEA, PBA to PHA, the degradation rate decreases because the crystallinity increases but the ester group density decreases. Such degradable polyurethane allows the antifoulants to release at a constant rate, where the release rate can be regulated by varying the polyester segments. Marine field tests reveal that the polyurethane-antifoulant system has good antifouling performance. Particularly, the degradable polyurethane exhibits excellent adhesion to the substrate ensuring the long duration of the system.

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1. Introduction

Marine biofouling is a global problem for marine industry and marine activities. Heavy metal-containing self-polishing copolymers (SPCs) coatings are effective to combat marine fouling. However, they were banned because they are ecologically harmful with release of high-toxic biocides and non-degradable main chains [1–4]. Degradable polymers can form a self-polishing surface and serve as carrier and release system of antifoulants. Particularly, with the degradable main chains, they exhibit good antifouling performance on both dynamic and static conditions, quite different from the traditional SPCs which have very limited antifouling ability in static marine environment [5–11].

Recently, polyesters with degradable backbone have been used for marine antibiofouling. However, the high crystallinity usually leads to poor adhesion to substrates and slow down the degradation, which limits their application [6,7,12]. It is necessary to decrease the crystallinity and spherulite size to improve the adhesion and regulate the degradation [5,8,13]. In the present study, we have prepared degradable polyurethanes with poly(ethylene adipate) (PEA), poly(1,4-butylen adipate) (PBA) or poly(1,6-hexamethylene adipate) (PHA) segments whose monomers are

made from renewable resources [14–18]. The polyurethane was used as the carrier of organic antifoulant (4,5-dichloro-2-octylisothiazolone). We have investigated the degradation of the polyurethanes, the release of antifoulants and the antifouling and mechanical properties of the systems. Our aim is to develop an eco-friendly marine antifouling system with good antifouling performance and high adhesion to the substrate.

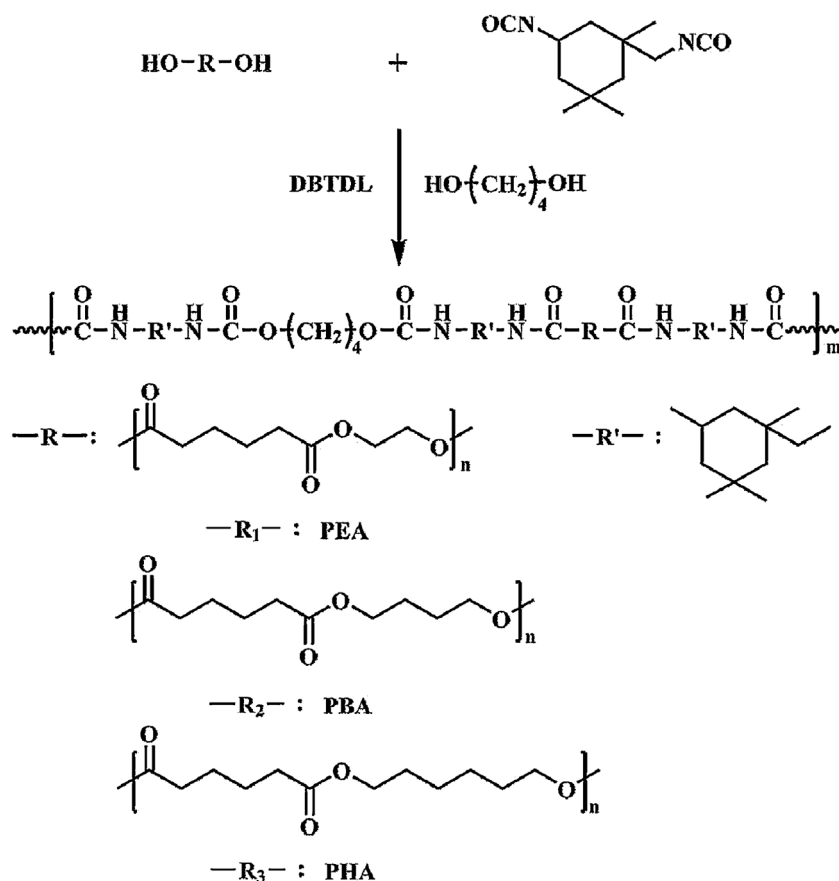
2. Experimental

2.1. Materials

Poly(ethylene adipate) diol (PEA, $M_w = 2000$ g/mol), poly(1,4-butylen adipate) diol (PBA, $M_w = 2000$ g/mol) and poly(1,6-hexamethylene adipate) diol (PHA, $M_w = 2000$ g/mol) from Xuchuan Chem were dried under reduced pressure for 2 h prior to use. 1,4-Butanediol (1,4-BD) from Aldrich was also dried in the same way. 3-isocyanatomethyl-3,5,5-trimethylcyclohexyl isocyanate (IPDI) from Aladdin and dibutyltin dilaurate (DBTDL) from Aldrich were used as received. Tetrahydrofuran (THF) was refluxed over CaH_2 and distilled prior to use. 4,5-Dichloro-2-octyl-isothiazolone (DCOIT) was kindly presented by Thankful Chemical Co. (Guangzhou). Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2013) [19]. Other reagents were used as received.

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Scheme 1. Synthesis of polyester based polyurethane.

Table 1
Characterization data of polyester based polyurethane.

Sample	<i>x</i> content (wt%) ^a	<i>M_n</i> × 10 ^{−4} (g/mol) ^b	Đ _M ^b	<i>T_g</i> ^c (°C)	<i>T_m</i> ^c (°C)	Δ <i>H_m</i> ^c (J/g)	<i>T</i> _{50%} ^d (°C)
PEA40-PU	39.3	13.7	1.32	−37.5	–	–	323
PEA60-PU	59.2	15.6	1.47	−33.1	–	–	341
PEA80-PU	79.1	14.4	1.42	−31.2	41.3	32.3	359
PBA80-PU	78.7	13.7	1.38	−44.6	44.6	36.3	357
PHA80-PU	78.6	13.5	1.39	–	49.8	45.1	348

^a Determined by ¹H NMR.^b Determined by GPC.^c Determined by DSC.

^d Temperature at 50% weight loss determined by TGA.

2.2. Synthesis of polyester-based polyurethane

Polyester-based polyurethane was synthesized by polyaddition (Scheme 1). First, IPDI reacted with polyester diol at 70 °C for 1 h in THF under nitrogen atmosphere, yielding a prepolymer. Subsequently, 1,4-BD and DBTDL were added as the chain extender and catalyst, respectively, and the mixture was allowed to react at 80 °C for 3 h. The product was precipitated into hexane twice, filtered, and dried under vacuum at 40 °C for 24 h. ¹H NMR (Fig. S1, 600 MHz, CDCl₃, ppm): 4.07 (OCH₂CH₂CH₂CH₂O), 3.79 (CONHCH(CH₂)₂), 3.21, 2.91 (C(CH₃)CH₂NHCO), 2.35 (COCH₂CH₂CH₂CH₂CO), FTIR (Fig. S2, cm⁻¹): 3530 (OH), 3380 (NH), 2950 (CH₂), 1735 (C=O), 1530 (CN). For convenience, the polyester-based polyurethanes are designated as PEAx-PU, PBAX-PU, PHAx-PU, where x represents the weight percentage of the polyester soft segment determined by ¹H NMR. The characterization data are summarized in Table 1 and the details can be found in the Supporting information (SI).

2.3. Polyester-based polyurethane characterization

¹H NMR spectra were recorded on a Bruker AV600 NMR spectrometer using CDCl₃ as solvent and tetramethylsilane (TMS) as internal standard.

FTIR spectra were recorded on a Bruker VECTOR-22 IR spectrometer. The spectra were collected at 64 scans with a spectral resolution of 4 cm^{-1} by KBr disk method.

The number-average molecular weights (M_n) and molecular weight dispersity (δ_M) were determined by GPC at 50 °C on a Waters 1515. A series of monodisperse polystyrene were used as standard in the fluent of DMF and LiBr with a flow rate of 1.0 ml/min.

TGA measurement was performed on a NETZSCH TG 209F1 instrument under nitrogen atmosphere at a heating rate of 10 °C/min in the range from 30 to 800 °C. The decomposition temperature ($T_{50\%}$) of PEA-PU increases with the content of PEA segments. PEA80-PU has the best thermal stability (Table 1 and Fig. S3 in SI).

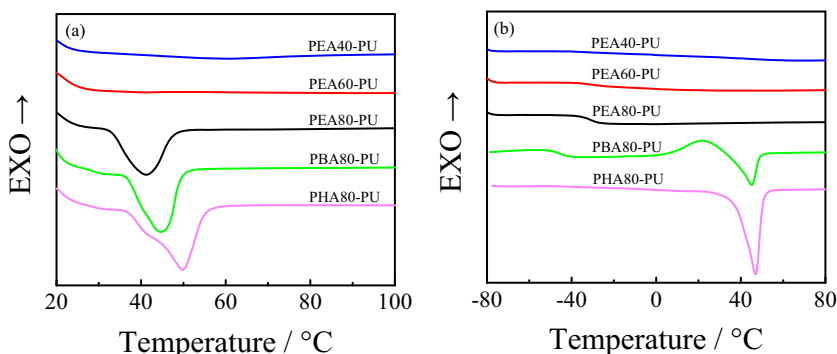


Fig. 1. (a) First and (b) second DSC thermograms of the polyurethanes.

DSC measurement was performed on a NETZSCH DSC 204F1 differential scanning calorimeter under a nitrogen flow of 50 ml/min. The sample was prepared by dropping a polyurethane solution in THF (30% w/v) on Teflon mold and dried at room temperature for 14 days to eliminate thermal history. Then, the sample (~ 5 mg) was heated to 100 °C at a rate of 10 °C/min and kept at 100 °C for 5 min and cooled to -80 °C at the same rate. The second scan thermogram was obtained by reheating it at 10 °C/min from -80 °C to 100 °C. The melting temperature (T_m) and melting enthalpy change (ΔH_m) were taken as the position of the peak and the area of melting endotherm in the first heating scan, respectively. The glass transition temperature (T_g) was taken as the midpoint of the transition during the second heating scan.

2.4. Preparation of polyurethane film

Each polymer film with dry thickness of ~ 400 μm was prepared via a solution-casting. Typically, a polyurethane solution in THF (30% w/v) was dropped on an epoxy resin panel and dried at room temperature for 14 days to remove the solvent for adhesion test, hydrolytic degradation test, antifoulant release rate measurement and marine field test.

2.5. Adhesion tests

The adhesion of the polyurethane to the epoxy resin panel was measured using pull-off adhesion tester (PosiTest AT-A Automatic) according to ASTM D4541-09 [20]. The measurement area is 20 mm in diameter and the pull rate was set at 0.2 MPa/s. Five different points on each sample were tested to obtain an average value.

2.6. Hydrolytic degradation test

The hydrolytic degradation test was performed on the polyurethane without antifoulant in ASW at 25 °C. The polyurethane film was coated on epoxy resin panel (20×20 mm²). Since the hydrolytic degradation occurs on the surface, we used the weighing method. The weighted sample (W_0) together with its panel was incubated in a tank of ASW that was renewed every two weeks. At a certain period of time, the panel was taken out, rinsed with deionized water three times, and dried in vacuum freeze dryer until constant weight (W_t) was obtained. The mass loss was designated as $(W_t - W_0)/\text{test area}$. For each sample, three coated panels were prepared and measured, and each data point was averaged over three successive and consistent measurements.

2.7. Determination of antifoulant release rate

DCOIT concentration was measured to determine its release rate by use of UV-vis spectrophotometer under a static condition

[21,22]. Typically, a composite film was coated on an epoxy resin panel (40×40 mm²) with the proportion of 90 wt% polyurethane and 10 wt% DCOIT, and then the panel was immersed in ASW. At a certain period of time, the panel was transferred from the holding tank to an individual measuring container. After 24 h immersion, 100 ml of seawater was taken from 250 ml of seawater in the individual measuring container into a 250 ml glass separating funnel, and it was extracted with 10 ml of hexane, and then the concentration of DCOIT was determined quantitatively by an Agilent Cary 60 UV-vis spectrophotometer at wavelength 290 nm. The release rate of DCOIT from the film was calculated from the DCOIT concentration in ASW using the following equation

$$R = (C \times 250 \times D)/A \quad (1)$$

where R is the release rate of DCOIT ($\mu\text{g cm}^{-2} \text{d}^{-1}$), C is the DCOIT concentration in ASW ($\mu\text{g ml}^{-1}$), D is the immersion time in the individual measuring container (one day) and A is the surface area of coated film (16 cm²). The cumulative release of DCOIT was calculated as follows

$$R_{0,x} = \sum \frac{(R_i + R_j)}{2} (j - i) \quad (2)$$

where $R_{0,x}$ is the cumulative release from Day 0 to Day x ($\mu\text{g cm}^{-2}$), i and j are time elapsed (days) since the start of the trial, for example, 0 and 1, 1 and 7, 7 and 21 days, and so forth, respectively. R_i and R_j are the release rates ($\mu\text{g cm}^{-2} \text{d}^{-1}$) of DCOIT from Day 0 to Day x , for example, on Days 0 and 1, Days 1 and 7, Days 7 and 21, and so forth, respectively, where Day 0 is included, the release rate on Day 0 (R_0) is taken as 0 $\mu\text{g cm}^{-2} \text{d}^{-1}$ of DCOIT.

2.8. Marine field tests

The field tests were performed from May 2014 to November 2014 at the inner Xiamen bay ($24^\circ 45' \text{N}$, $118^\circ 07' \text{E}$) in China, where the temperature of surface water ranges from 12.5 to 29.5 °C and the salinity is from 21.2‰ to 30.5‰. The samples applied onto the glass fiber reinforced epoxy resin panels ($250 \times 100 \times 3$ mm³) were lowered into seawater at a depth of 0.2–2.0 m. After a certain period of time, the panels were taken out of the sea, carefully washed with seawater and photographed, and then placed back in the sea to continue the test. The antifouling activity was evaluated according to ASTM D6990-05 (2011) [23]. The range of the fouling rating (FR) was 0–100. FR for a coating free from biofouling settlement was recorded as 100. FR of each coating was obtained by deducting from 100, depending on the percentage of area covered by macrofouling, that is, the fouling rating essentially reflects the non-fouled area. There, the biofouling pressure mainly comes from barnacles, algae and slime. Biofouling attachment within 1 cm from the edge was not counted.

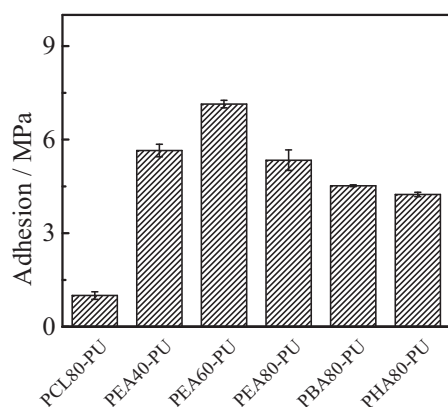


Fig. 2. Adhesion strength of the polyurethane to epoxy panel.

3. Results and discussion

DSC thermograms of polyester based polyurethanes are presented in Fig. 1. In comparison with PEA40-PU and PEA60-PU without melting endothermic peak, PEA80-PU, PBA80-PU and PHA80-PU show the melting peaks at 41.3, 44.6 and 49.8 °C on the first heating thermogram, respectively (Fig. 1a). However, PEA80-PU no longer exhibits melting peak on the second heating thermogram after cooling to −80 °C at a rate of 10 °C/min (Fig. 1b), indicating that it does not crystallize from the melt upon cooling. Considering that solution-casting coating instead of melt-casting is used in industry, we are more interested in the first heating results. Clearly, PEA40-PU and PEA60-PU are amorphous. PEA80-PU has the smallest enthalpy change ($\Delta H_m = 32.3$ J/mol) from the melting peak area in comparison with PBA80-PU ($\Delta H_m = 36.3$ J/mol) and PHA80-PU ($\Delta H_m = 45.1$ J/mol), indicating the crystallinity of PEA80-PU has the lowest crystallinity, which is conducive to the degradation.

We measured the adhesion of the polyurethane to the substrate, which is critical for applications of marine antifouling materials. Fig. 2 shows the adhesion strength between epoxy resin substrate and the polyurethane. We used the biodegradable poly(ϵ -caprolactone)-based polyurethane (PCL80-PU) as the reference material. It only has an adhesion value of 1.0 ± 0.21 MPa. However, the polyester based polyurethane in this study has excellent adhesion property with adhesion strength above 4.0 MPa. For PEA40-PU, PEA60-PU and PEA80-PU, the adhesion strength values are 5.7, 7.2, and 5.3 MPa, respectively. Namely, PEA60-PU has the highest adhesion strength. As we know, both PEA60-PU and PEA40-PU are amorphous, but the former with more soft segments has more contacts with the substrate surface with more hydrogen bonds formed between them. However, further increasing the soft segment content increases up to 80 wt% leads the polyurethane to be semicrystalline, which sharply reduces its contacts with the substrate. That is why PEA60-PU exhibits the lowest adhesion strength.

We studied the hydrolytic degradation of the polyurethanes in ASW at 25 °C as function of time (Fig. 3). Generally, each film exhibits a relatively rapid mass loss at the initial stage. This is probably caused by a quick release of some low molar mass components [24,25]. Seven days later, the mass loss is almost linear with time, indicating a degradation with a constant rate. For PEA-PU, as the soft segment content increases from 40 wt% to 80 wt%, the mass loss increases. On the other hand, either PBA80-PU or PHA80-PU has lower degradation rate than PEA80-PU. As we know, the degradation rate is profoundly influenced by the crystallinity of the polymer. Since the amorphous region is easier for small molecules to penetrate, the degradation rate increases as the crystallinity decreases. On the other hand, the degradation rate increases with the ester group density in the polymer [5,26]. Thus, although

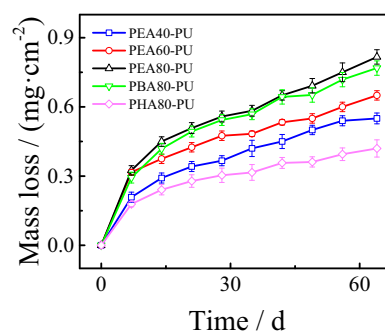


Fig. 3. Time dependence of mass loss of the polyurethane in ASW at 25 °C.

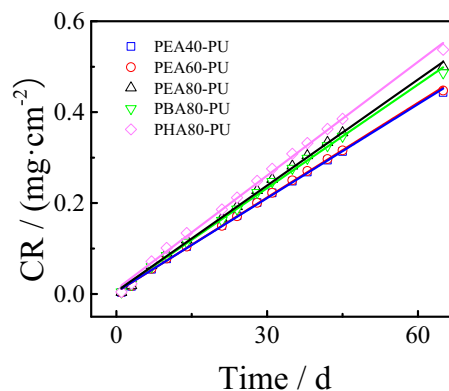


Fig. 4. Time dependence of the cumulative release (CR) of DCOIT from the polyurethane.

PEA40-PU and PEA60-PU are amorphous, they still exhibit lower degradation rates than PEA80-PU with the highest ester group density. Particularly, PEA80-PU has a higher degradation rate than either PBA80-PU or PHA80-PU because it has the highest ester group density and lowest crystallinity.

The duration and antifouling performance of an anti-biofouling system have much dependence on the controlled release of antifoulants. If the antifoulants are released too quickly, the system would have an inevitably short service life. On the contrary, the antifouling efficiency would be limited with a low antifoulants release rate. If the release rate is too high in the initial stage and then exponentially decreases with time, the system does not efficient and prolonged performance. The cumulative release of DCOIT from the polyurethane in a certain period of time is depicted in Fig. 4. For each film, the amount of released DCOIT is linearly related to time with a correlation coefficient >0.99 , indicating the release has a constant rate. Since DCOIT is very hydrophobic with a solubility of 5 ppm in water [6,27], it can be uniformly dispersed in the polymer and does not readily diffuse into seawater. So the release is largely determined by the polymer degradation but not the diffusion. As the polymer degrades at a constant rate, DCOIT is gradually released with a constant rate. Note that the slope for PEA80-PU is higher than that for either PEA60-PU or PEA40-PU because it has the highest degradation rate with the highest DCOIT release rate. PBA80-PU has a slower DCOIT release than PEA80-PU because the former has a lower degradation rate. Among PEA80-PU, PBA80-PU and PHA80-PU, PHA80-PU has the highest release rate though it has the lowest degradation rate. This is probably because the high crystallinity leads to poor compatibility and weak interaction between PHA80-PU and DCOIT, which is favorable to the release of antifoulants.

The antifouling performance of the systems was evaluated by marine field tests. The typical images of tested panels after immer-

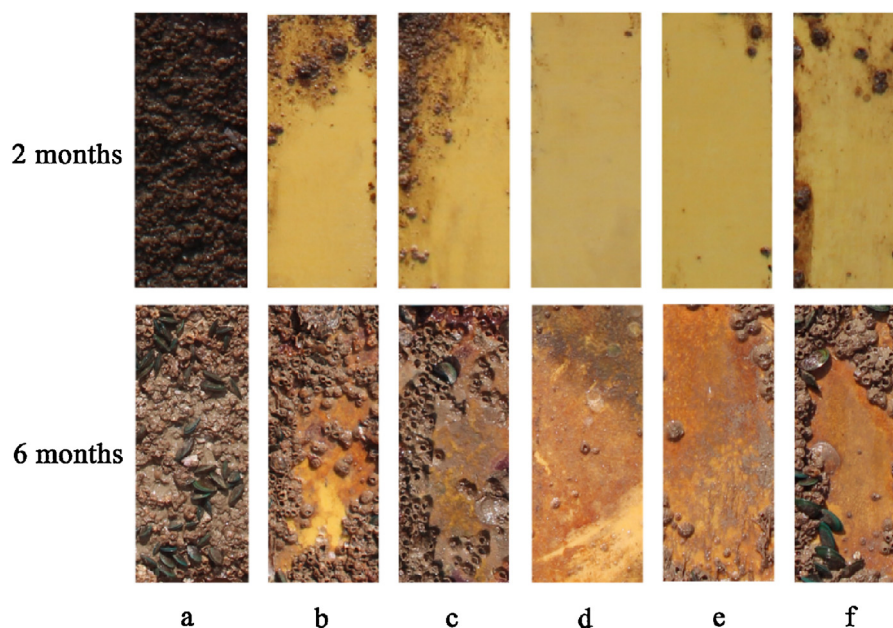


Fig. 5. Images of panels after immersion in the seawater (May–November 2014, Xiamen, China). (a) Control panel without coating. (b) Panel coated with PEA40-PU/DCOIT of 90/10. (c) Panel coated with PEA60-PU/DCOIT of 90/10. (d) Panel coated with PEA80-PU/DCOIT of 90/10. (e) Panel coated with PBA80-PU/DCOIT of 90/10. (f) Panel coated with PHA80-PU/DCOIT of 90/10.

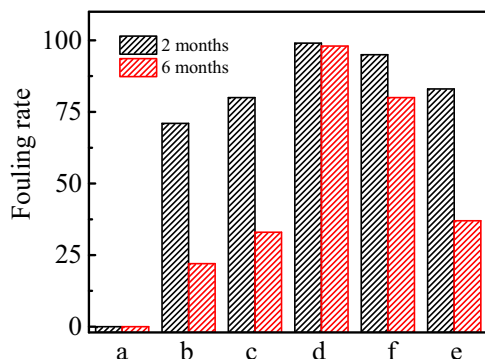


Fig. 6. Fouling rating after immersion in the seawater. (a) Control panel without coating. (b) Panel coated with PEA40-PU/DCOIT of 90/10. (c) Panel coated with PEA60-PU/DCOIT of 90/10. (d) Panel coated with PEA80-PU/DCOIT of 90/10. (e) Panel coated with PBA80-PU/DCOIT of 90/10. (f) Panel coated with PHA80-PU/DCOIT of 90/10.

sion in seawater are presented in Fig. 5 and the corresponding fouling rating is shown in Fig. 6. After 2 months, the control panel with epoxy surface is completely covered by the biofouling, indicating a heavy fouling pressure in the testing site. However, much less fouling is observed on the surface constructed by a degradable polymer containing DCOIT and the *in situ* antifouling performance increases with the content of the degradable segments (from PEA40-PU to PEA80-PU) but slightly decreases as the segments change from PEA, PBA to PHA. This is because a degradable polymer can form a self-polishing surface to polish the attached living organisms or inorganics away. Moreover, it can release DCOIT to inhibit the biofouling. The improved antifouling performance from PEA40-PU to PEA80-PU can be attributed to the increased erosion rate and release rate of DCOIT (Figs. 3 and 4). Note that either PEA80-PU/DCOIT or PBA80-PU/DCOIT exhibits good antifouling performance with a few organisms colonized, and the former's is even better. This is because they have higher degradation rate and DCOIT release rate. For PHA80-PU/DCOIT, it is fouled

because of the low self-polishing rate. Actually, both a reasonable self-polishing rate and a controlled release of antifoulants are critical for good anti-biofouling ability of a system. We also evaluated their long-term antifouling ability for 6 months. The panel coated with PEA40-PU/DCOIT or PEA60-PU/DCOIT was fouled by some marine organisms because the system with a slow degradation and a slow DCOIT release has limited antifouling ability. The PEA80-PU/DCOIT or PBA80-PU/DCOIT exhibits good antifouling performance but the PHA80-PU/DCOIT is fouled. This fact further indicates the *in situ* antifouling efficiency of the coatings is determined by both the erosion rate and the DCOIT release rate. It should be noted that we only examined the antifouling performance under static conditions. The system is expected to exhibit better antifouling performance on dynamic conditions because the degradation rate and the antifoulants release rate increase upon water flows.

4. Conclusion

We have prepared polyester based polyurethane and used it as carrier and release system of antifoulants. The system exhibits excellent adhesion to the substrate. The degradation rate can be regulated by varying the degradable segment content and the ester group density. The antifoulants are released at a constant rate, which can be controlled by the degradation rate. The system with appropriate degradation rate and antifoulant release rate exhibits good antifouling performance and is expected to find applied in marine antifouling.

Acknowledgments

The financial support of Ministry of Science and Technology of China (2012CB933802), National Natural Science Foundation of China (51303059, 51573061 and 51673074), China Postdoctoral Science Foundation (2016M590777), Hong Kong Scholars Program (XJ2014020) is acknowledged.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.porgcoat.2016.12.011>.

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